

ENTHUSIAST ADVANCE COURSE

PHASE : MEF-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 04-03-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	3	4	2	3	3	1	3	4	2	1	1	1	3	2	1	2	2	1	2	1	1	2	3	1	2	1	4	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	1	1	1	2	1	1	1	4	2	2	4	2	1	3	3	2	1	1	1	1	2	2	2	3	1	2	3	4	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	4	4	3	1	3	4	1	1	1	3	3	1	1	3	2	1	1	3	4	2	2	1	3	2	2	4	3	4	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	2	3	2	4	4	3	1	3	4	3	1	3	1	1	2	3	1	4	2	3	4	4	4	3	4	1	4	1	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	3	3	4	3	4	1	1	2	2	3	2	4	1	2	3	1	1	1	1	4	3	4	2	2	3	2	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	3	3	3	3	4	4	2	1	3	4	2	3	2	3	3	4	2	3	3	2	3	1	3	2	1	4	1	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	2	2	2	4	3	4	3	3	2	4	2	2	3	4	1	4	3	1	3										

HINT - SHEET**SUBJECT : PHYSICS****SECTION - A**1. **Ans (3)**

$$q = ne$$

$$80 \times 10^{-6} = n \times 1.6 \times 10^{-19}$$

2. **Ans (3)**

If both charges are similar then neutral point will lie between them.

3. **Ans (4)**

Let r be the distance between q_1 and q_2 .

According to Coulomb's law, the force between them is,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q_1(q - q_1)}{r^2} \quad (\because q_1 + q_2 = q)$$

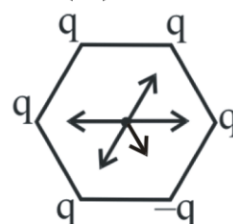
For F to be maximum $\frac{dF}{dq_1} = 0$ as q and r are

constants.

$$\therefore \frac{d}{dq_1} \left[\frac{1}{4\pi\epsilon_0 r^2} (q_1 q - q_1^2) \right] = 0$$

$$q - 2q_1 = 0 \quad \left[\text{As } \frac{1}{4\pi\epsilon_0 r^2} \neq 0 \right]$$

$$\frac{q_1}{q} = \frac{1}{2} = 0.5$$

4. **Ans (2)**

$$F_{\text{net}} \neq 0$$

5. **Ans (3)**

$$\vec{E} \text{ at centre} = 0$$

Because of symmetry when one charge is removed

$$\vec{E} = \frac{kq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

($\because \vec{E}$ due to 10 charges will cancel and out each other)

$\therefore E_{\text{net}}$ will be only because of 11th charge)

6. **Ans (3)**

$$F \propto q_1 q_2$$

$$\Rightarrow \frac{F}{F'} = \frac{1}{1.1 \times 0.9}$$

$$F' \propto (1.1q_1)(0.9q_2)$$

$$\Rightarrow F' = 1.1 \times 0.9 \times 100$$

$$= 0.99 \times 100$$

$$= 99 \text{ N}$$

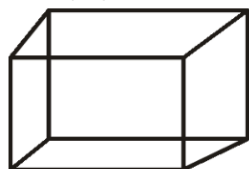
10. **Ans (2)**

$$E = \frac{kq}{1^2} + \frac{kq}{2^2} + \frac{kq}{4^2} + \frac{kq}{8^4} \dots$$

$$= kq \left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} \dots \infty \right)$$

$$= kq \left(\frac{1}{1 - \frac{1}{4}} \right) = \frac{4}{3} kq$$

12. **Ans (1)**



$$\phi = \frac{2}{\epsilon_0} + \left(\frac{1}{4} \left(\frac{1}{\epsilon_0} \right) \times 12 \right)$$

$$= \frac{5}{\epsilon_0}$$

14. **Ans (3)**

$$\frac{kQx_1}{R^3} = \frac{kQ}{x_2^2}$$

$$R^3 = x_1 x_2^2$$

16. **Ans (1)**

$$\frac{F_e}{F_G} = \frac{\frac{Kq_1 q_2}{r^2}}{\frac{Gm_1 m_2}{r^2}}$$

$$2.4 \times 10^{39} = \frac{K}{G} \times \frac{(1.6 \times 10^{-19})^2}{(9.11 \times 10^{-31} \times 1.67 \times 10^{-27})}$$

$$\frac{K}{G} = \frac{2.4 \times 10^{39} \times 15.2137 \times 10^{-58}}{2.56 \times 10^{-38}}$$

$$= 14.26 \times 10^{19}$$

$$= 1.426 \times 10^{20}$$

$$\gg 10^{20}$$

17. **Ans (2)**

$$V \propto \frac{Q}{R} \text{ So } Q \propto R \text{ at constant } V$$

$$\text{So } \frac{q_1}{q_2} = \frac{R_1}{R_2}$$

19. **Ans (1)**

$$U = V_{p,m}$$

$$U = - \left[\frac{GM}{2R} + \frac{GM}{2R} \right] \cdot m$$

$$U = \frac{-GMm}{R}$$

25. **Ans (1)**

Use MECL

$$\frac{-GMm}{R} + \frac{1}{2} m \frac{gR}{2} = \frac{-GMm}{R + \frac{R}{4}} + \frac{1}{2} m V_p^2$$

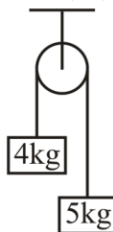
26. **Ans (2)**

$$\text{Acceleration} = \frac{10^5}{(10)(1500)} = 6.66$$

$$\text{Force} = (4)(1500)(6.66)$$

$$= 40,000 \text{ N}$$

27. **Ans (1)**



$$T = \frac{2m_1 m_2}{m_1 + m_2} g$$

$$T = \frac{(2)(4)(5)}{(5+4)} g$$

$$T = \frac{40}{9} g = \frac{400}{9} \text{ N}$$

28. Ans (4)

$$\frac{mg}{m(g-a)} = \frac{5}{4} \Rightarrow 4g = 5g - 5a$$

$$5a = g$$

$$a = \frac{g}{5}$$

29. Ans (3)

$$6g = (3+m)g \sin 37^\circ$$

$$6 = (3+m) \left(\frac{3}{5} \right)$$

$$10 = 3 + m$$

$$m = 7 \text{ kg}$$

30. Ans (2)

$$a = \frac{m_2 g}{m_1 + m_2} = \frac{(3)(10)}{(3+2)} = 6 \text{ m/s}^2$$

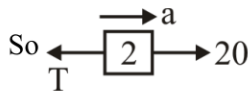
$$S = 0 + \frac{1}{2} a t^2$$

$$S = \left(\frac{1}{2} \right) (6) (2)^2 = 12 \text{ m}$$

$$S = 12 \text{ m}$$

31. Ans (1)

$$\text{Acceleration} = \frac{F_{\text{net}}}{m} = \frac{20-5}{7} = \frac{15}{7}$$



$$20 - T = 2a$$

$$T = \frac{110}{7} \text{ N}$$

32. Ans (1)

Reading = tension in spring

33. Ans (1)

$$\text{Acceleration} = \frac{24}{8} = 3 \text{ m/s}^2$$

$$T = (3)(6) = 18 \text{ N}$$

34. Ans (1)

$$\text{Acceleration} = \frac{F_{\text{net}}}{\text{mass}} = \frac{144 - 120}{12} = 2 \text{ m/s}^2$$

35. Ans (2)

$$\vec{F} = 4\hat{i} + 4\hat{j} + 7\hat{k}$$

$$\text{then } |\vec{F}| = \sqrt{4^2 + 4^2 + 7^2} = \sqrt{81} = 9 \text{ N}$$

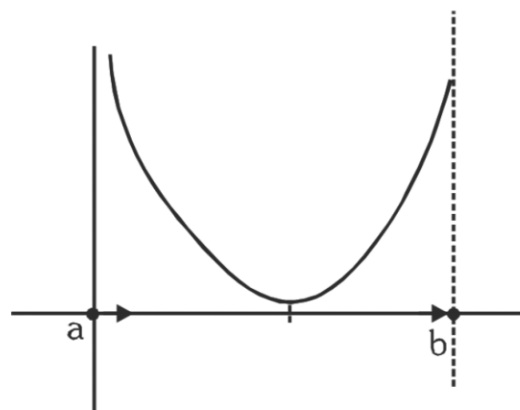
$$\text{Acceleration} = 3 \text{ m/s}^2 \text{ so mass } \frac{F}{a} = \frac{9}{3} = 3$$

SECTION - B

36. Ans (1)

$$\epsilon_0 = \frac{Q_1 Q_2}{R \cdot r^2} = \frac{(AT)^2}{(MLT^{-2}) \cdot L^2} = [M^{-1} L^{-3} T^4 \cdot A^2]$$

37. Ans (1)



$a \rightarrow +ve$

$b \rightarrow -ve$

38. Ans (1)

We have centripetal force equation

$$q \left(\frac{2k\lambda}{r} \right) = \frac{mv^2}{r}$$

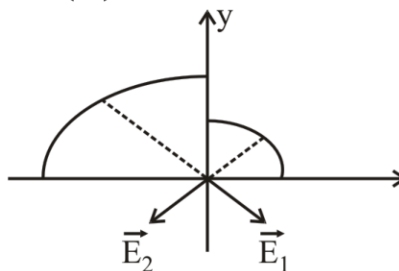
$$\text{so } v = \sqrt{\frac{2kq\lambda}{m}}$$

Now,

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$$

$$\text{where } k = \frac{1}{4\pi\epsilon_0}$$

40. Ans (2)



$$E_{\text{net}} = \sqrt{E_1^2 + E_2^2}$$

$$= \sqrt{\left(\frac{\sqrt{2}k\lambda}{2R} \right)^2 + \left(\frac{\sqrt{2}k\lambda}{R} \right)^2}$$

$$= \frac{\sqrt{2}k\lambda}{R} \sqrt{\frac{1}{4} + 1} = \frac{\sqrt{10}}{2} \frac{k\lambda}{R}$$

43. Ans (2)

For a conducting sphere

$$E = \frac{\sigma}{\epsilon_0}$$

$$V = \frac{\sigma R}{\epsilon_0}$$

as both spheres have same potential after connecting with wire,

$$V_1 = V_2$$

$$\sigma_1 R_1 = \sigma_2 R_2$$

$$\Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$$

45. Ans (3)

The line joining the sun to the planet sweeps out equal areas in equal time interval i.e. areal velocity is constant.

$$\frac{dA}{dt} = \text{constant or } \frac{A_1}{t_1} = \frac{A_2}{t_2}$$

where, A_1 = area under SCD

A_2 = area under SAB

$$\Rightarrow t_1 = \frac{A_1}{A_2} t_2$$

Given, $A_1 = 2A_2$

$$\therefore t_1 = 2t_2$$

48. Ans (1)

$$\text{Acceleration} = \frac{80}{m} \text{ mass till point P} = \left(\frac{m}{10}\right) 7$$

Hence force = tension = ma

$$\Rightarrow \left(\frac{80}{m}\right) \left(\frac{m}{10}\right) 7 = 56 \text{ N}$$

49. Ans (1)

$$\text{Using } \Sigma \vec{T} \cdot \vec{a} = 0$$

$$4Ta_2 = Ta_1$$

$$a_1 = 4a_2$$

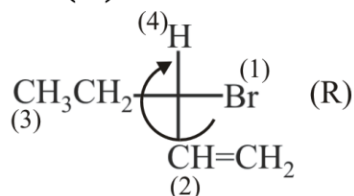
SUBJECT : CHEMISTRY

SECTION-A

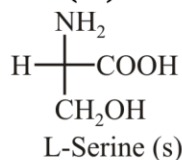
55. Ans (3)

Two different groups at ortho position.

56. Ans (1)



57. Ans (2)



L-Serine (S)

58. Ans (3)

Homomers because both (A) and (B) have same R/S configuration.

78. Ans (1)

Temp coefficient μ is ratio of rate constants at temp difference 10°C

80. Ans (4)

Catalyst decreases the value of activation energy

81. Ans (2)

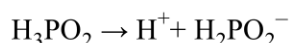
Unit of k = unit of A for 1st order $rx^n = s^{-1}$

82. Ans (2)

conc. term including volume will be temp. dependent

83. Ans (1)

$N = M \times n$ - factor



n factor = 1

84. Ans (3)



$$0.01V \quad 0.01V \quad 0.01V$$

$$[NaCl] = \frac{0.01V}{V + V} = 0.005M$$

85. Ans (2)

Adding catalyst ΔH remains same

SECTION-B

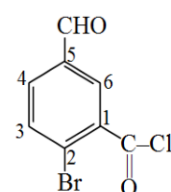
86. Ans (2)

• Benzene ring must be selected as parent chain

• $-\text{C}-\text{Cl}$ group is principle functional group.

• $-\text{CH}=\text{O}$, $-\text{Br}$, are side chain.

• Numbering of benzene ring according to least locant number rule.



2-bromo-5-formyl benzene carbonylchloride

97. **Ans (3)**

For zero order reaction rate of reaction remains constant -

98. **Ans (1)**

$$m = \left(\frac{1000 \times M}{1000 \times d - M \times M_w} \right)$$

$M = 2$

99. **Ans (3)**

$$kt = \ln \left(\frac{A_0}{A_t} \right)$$

$$t_{90\%} = \frac{1}{k} \ln \left(\frac{100}{10} \right) = \frac{1}{k} \ln(10)$$

SUBJECT : BIOLOGY-I

SECTION-A

105. **Ans (1)**

NCERT-XII, Pg#182, Para-9.1.2

108. **Ans (1)**

NCERT-XII Pg. # 54/51(H) Para: 3.7

112. **Ans (4)**

NCERT (XII) Pg. # 54

113. **Ans (4)**

NCERT XII, Pg # 52

121. **Ans (2)**

NCERT-XII, Pg. # 49

Module-I, Pg.# 26

124. **Ans (3)**

NCERT-XII, Pg. # 50

129. **Ans (1)**

Module Pg. # 85

130. **Ans (2)**

Module Pg. # 22

135. **Ans (1)**

NCERT-XII, Pg. # 50

SECTION-B

136. **Ans (2)**

NCERT(XII) Pg#168 Para:9.1.2

137. **Ans (3)**

NCERT(XII) Pg#168 Para:9.1.2

138. **Ans (1)**

NCERT (XIIth) Pg. # 167, 168,169, Para 9.1.2

143. **Ans (3)**

NCERT Pg. # 43

144. **Ans (4)**

NCERT (XII) Pg. # 53

149. **Ans (3)**

Module-I, Pg. # 45

150. **Ans (4)**

Module-01, Pg. # 87

SUBJECT : BIOLOGY-II

SECTION-A

151. **Ans (3)**

NCERT (XII) Pg. # 153, Para. # 8.2.5

155. **Ans (3)**

NCERT Pg. # 149 (E)

156. **Ans (3)**

NCERT (XIIth) Pg. # 63

157. **Ans (4)**

NCERT-XII, Pg No. # 159

159. **Ans (2)**

NCERT-XII, Pg. # 127

162. **Ans (4)**

NCERT (XII) Pg # 128/139(H) Para : 7.2

164. **Ans (3)**

NCERT pg.# 126, line 1

SECTION-B

187. **Ans (4)**

NCERT (XII) Pg. # 147,148,149